

5(a)(i)	(mean hind limb length) should decrease / legs get shorter ;	1
5(a)(ii)	(mean hind limb length) will vary ;	1
5(b)	<i>any five from descriptions (D), suggestions (S) and explanations (E):</i> D1 on, all / seven / the, experimental / founder, islands hind limb length decreased ; S1 <u>directional selection</u> ; D2 on source island hind limb length stayed the same ; S2 <u>stabilising selection</u> ; <i>explanation in context of experimental islands:</i> E1 <u>selection pressure</u> is, (reduced / thinner) perch / branch, (diameter); E2 short (hind) limbs are, a <u>selective</u> advantage / <u>selected</u> for, on, experimental islands / founder islands / thin branches ; E3 (short hind limbs) allow / increase, stability / survival ; E4 (survivors), reproduce / breed / pass on allele(s) (for short hind limbs) ; E5 <u>allele</u>(s) for short (hind) limbs increases in frequency ;	5

Question	Answer	Marks
5(c)	any three from: 1 <u>allopatric</u> ; 2 populations / lizards / <i>A. sagrei</i>, on different islands, geographically / geologically, isolated / separated ; 3 have different, selection pressures / environmental conditions ; 4 different mutations occur ; 5 <u>genetic drift</u> ; 6 no gene flow / reproductive isolation, between, lizards on different islands / populations / (new) species ; 7 genetic differences accumulate ;	3
5(d)	any three from: 1 (each island has a) small population ; 2 with, low genetic diversity / small gene pool ; 3 inbreeding depression / low heterozygosity / high homozygosity ; 4 environmental change / natural disaster, may, kill them all / make (whole / each) population extinct ;	3